

ANSWER KEY — Integration Lesson 21 Test: Completing the Power Rule

For the grader · full working shown · give credit for correct method with small arithmetic slips · missing "+ C" on an indefinite integral costs the point only once per test

1. Rewrite as a power of x , then integrate: $\int 1/x^3 \, dx$. Check by differentiating.

Work: $1/x^3 = x^{-3}$. New power: $-3 + 1 = -2$. So $x^{-2}/(-2) = -1/(2x^2)$. Check: $d/dx[-1/2x^{-2}] = -1/2 \cdot (-2)x^{-3} = x^{-3} \checkmark$. **Answer:** $-1/(2x^2) + C$. (Classic error: dividing by the OLD power -3 , giving $x^{-2}/(-3)$ — the differentiation check exposes it.)

2. Integrate: $\int x^{1/4} \, dx$.

Work: new power: $1/4 + 1 = 5/4$. Divide by $5/4 =$ multiply by $4/5$: $x^{5/4}/(5/4) = (4/5)x^{5/4}$. Check: $(4/5)(5/4)x^{1/4} = x^{1/4} \checkmark$. **Answer:** $(4/5)x^{5/4} + C$. (Classic error: writing $(5/4)x^{5/4}$ — multiplying instead of dividing by the new power.)

3. Integrate: $\int 12/x^4 \, dx$.

Work: $12/x^4 = 12x^{-4}$. Then $12 \cdot x^{-3}/(-3) = -4x^{-3} = -4/x^3$. Check: $d/dx[-4x^{-3}] = -4 \cdot (-3)x^{-4} = 12x^{-4} \checkmark$. **Answer:** $-4/x^3 + C$. (Watch for a dropped minus sign — the most common slip with negative powers.)

4. Integrate: $\int (\sqrt{x} + 4) \, dx$.

Work: term by term: $\int x^{1/2} \, dx = x^{3/2}/(3/2) = (2/3)x^{3/2}$; $\int 4 \, dx = 4x$. Check: $(2/3)(3/2)x^{1/2} + 4 = \sqrt{x} + 4 \checkmark$. **Answer:** $(2/3)x^{3/2} + 4x + C$. (Classic error: forgetting the constant term integrates to $4x$, not 4 .)

5. Split the fraction, then integrate: $\int (x^4 + 5)/x^2 \, dx$.

Work: $(x^4 + 5)/x^2 = x^4/x^2 + 5/x^2 = x^2 + 5x^{-2}$. Then $x^3/3 + 5 \cdot x^{-1}/(-1) = x^3/3 - 5/x$. Check: $d/dx[x^3/3 - 5x^{-1}] = x^2 + 5x^{-2} \checkmark$. **Answer:** $x^3/3 - 5/x + C$. (Grader note: trying to integrate top and bottom separately — $x^5/5$ over $x^3/3$ — is a conceptual error worth discussing; there is no "quotient rule" for integrals.)

6. Evaluate exactly: $\int_1^3 2/x^2 \, dx$.

Work: antiderivative of $2x^{-2}$ is $2 \cdot x^{-1}/(-1) = -2/x$. Then $[-2/x]_1^3 = (-2/3) - (-2/1) = -2/3 + 2 = 4/3$. **Answer:** $4/3$. (Classic error: mangling the double negative to get $-8/3$ or $2/3$ — insist on writing the subtraction out.)

7. Evaluate exactly: $\int_0^9 \sqrt{x} \, dx$.

Work: antiderivative $(2/3)x^{3/2}$. At 9: $9^{3/2} = (\sqrt{9})^3 = 27$, so $(2/3) \cdot 27 = 18$. At 0: 0. Difference: $18 - 0$. **Answer: 18.** (Classic error: computing $9^{3/2}$ as $9 \cdot 1.5 = 13.5$ — remind them it means $(\sqrt{9})^3$.)

8. Evaluate exactly: $\int_4^9 1/\sqrt{x} \, dx$.

Work: $1/\sqrt{x} = x^{-1/2}$; antiderivative $x^{1/2}/(1/2) = 2\sqrt{x}$. Then $[2\sqrt{x}]_4^9 = 2 \cdot 3 - 2 \cdot 2 = 6 - 4 = 2$.

Answer: 2. (Sanity check for grader: region is 5 wide, heights between $1/3$ and $1/2$, so answer must be between $5/3$ and $5/2$ — 2 fits.)

9. A student writes $\int 1/x^2 \, dx = \ln(x^2) + C$. Explain the error and give the correct answer.

Work: $\ln$ is ONLY the antiderivative of $1/x$ (the power -1 , where the power rule divides by zero).

For any other power — here $1/x^2 = x^{-2}$ — the power rule works fine: $x^{-1}/(-1) = -1/x$. Check:

$d/dx[-1/x] = 1/x^2$ ✓, while $d/dx[\ln(x^2)] = 2/x$ ✗. **Answer: $-1/x + C$; $\ln$ is reserved for the single power $n = -1$.**

10. Sap flows at $r(t) = 6/\sqrt{t}$ liters per hour, t from 1 to 4 hours. (a) Rewrite $r(t)$ as a power of t . (b) Total liters collected.

Work: (a) $r(t) = 6t^{-1/2}$. (b) Total = $\int_1^4 6t^{-1/2} \, dt$; antiderivative $6 \cdot 2\sqrt{t} = 12\sqrt{t}$ (check: $d/dt[12t^{1/2}] = 6t^{-1/2}$ ✓). $[12\sqrt{t}]_1^4 = 12 \cdot 2 - 12 \cdot 1 = 24 - 12 = 12$. Units: $(L/hr) \cdot hr = \text{liters}$. **Answer: (a) $6t^{-1/2}$ (b) 12 liters.**

(Award (a) even if (b) fails; award (b) method credit if the antiderivative $12\sqrt{t}$ is right but arithmetic slips.)